Data Analytics Project: IGN Game Reviews Analysis

Dataset Used: ign.csv

This document outlines a structured, end-to-end data analysis project based on game reviews from the IGN dataset. The project is divided into five scenarios, increasing in difficulty from basic data handling to advanced analysis and visualization, ensuring comprehensive skill coverage.

Project Goal: To analyse game review scores, platform popularity, and genre distribution over time using the `ign.csv` dataset.

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

🎯 Project Coverage

Category	Skills Covered
Data Pre-processing	Data Loading, Missing Value Handling (Mean/Mode), Type Conversion, Column Removal
Pandas Operations	Grouping, Filtering, Aggregation, Feature Engineering (score_category, editors_choice conversion)
NumPy Calculations	Array conversion, Yearly score growth calculation (np.diff)
Visualization	Line Chart, Bar Chart, Pie Chart, Histogram, Stacked Bar Chart, Saving Plots

Scenario 1: Basic Data Loading & Cleaning (□ Easy)

Description:

This scenario is mainly focussing on how to load the data and also cleaning the IGN game reviews using pandas. The main goal of this scenario to display first and last rows, removing un-necessary columns, identifying missing values and also preparing the dataset for analysis. This ensures that dataset is clean and also ready for the future visualization

Tasks	Description
1	Load the `ign.csv` dataset using Pandas.
2	Display the first 5 rows (`head()`), last 5 rows (`tail()`), and the shape of the dataset.
3	Remove the unnecessary column: `"Unnamed: 0"`.
4	Check for missing values in `score`, `genre`, and `platform`.
5	Handle missing values: Fill the numeric column `score` with the mean, and the categorical column `genre` with the mode.
6	Ensure correct data types: Convert `score` → float and `release_year`, `release_month`, `release_day` → integer.

```
------------ RESTART: C:\pythontasks\Data_Analytics\Project-2\main.py ------------
   Unnamed: 0 score_phrase  ... release_month release_day
0          0       Amazing  ...             9          12
1          1       Amazing  ...             9          12
2          2         Great  ...             9          12
3          3         Great  ...             9          11
4          4         Great  ...             9          11

[5 rows x 11 columns]
       Unnamed: 0 score_phrase  ... release_month release_day
18620       18620         Good  ...             6          29
18621       18621       Amazing  ...             6          29
18622       18622     Mediocre  ...             6          28
18623       18623  Masterpiece  ...             6          28
18624       18624  Masterpiece  ...             6          28

[5 rows x 11 columns]
(18625, 11)
unnecessary columns:       score_phrase  ... release_day
0           Amazing  ...           12
1           Amazing  ...           12
2             Great  ...           12
3             Great  ...           11
4             Great  ...           11
...             ...  ...          ...
18620          Good  ...           29
18621       Amazing  ...           29
18622      Mediocre  ...           28
18623  Masterpiece  ...           28
18624  Masterpiece  ...           28

[18625 rows x 10 columns]
missing values:
 score         0
genre         36
platform       0
dtype: int64
```

```
dtype: int64
handling missing values:    score       genre
0    9.0  Platformer
1    9.0  Platformer
2    8.5      Puzzle
3    8.5      Sports
4    8.5      Sports
       score  release_year  release_month  release_day
0        9.0          2012              9           12
1        9.0          2012              9           12
2        8.5          2012              9           12
3        8.5          2012              9           11
4        8.5          2012              9           11
...      ...           ...            ...          ...
18620    7.6          2016              6           29
18621    9.0          2016              6           29
18622    5.8          2016              6           28
18623   10.0          2016              6           28
18624   10.0          2016              6           28
```

Conclusion:

In this scenario, IGN dataset was successfully loaded and cleaned using pandas. Missing values were identified and handles correctly using mean and mode operations. An index column was removed to improve data structure. This steps ensures that dataset is ready for the future analysis and also visualization.

Scenario 2: Line Graph (Score Trend) + Save (□ Easy)

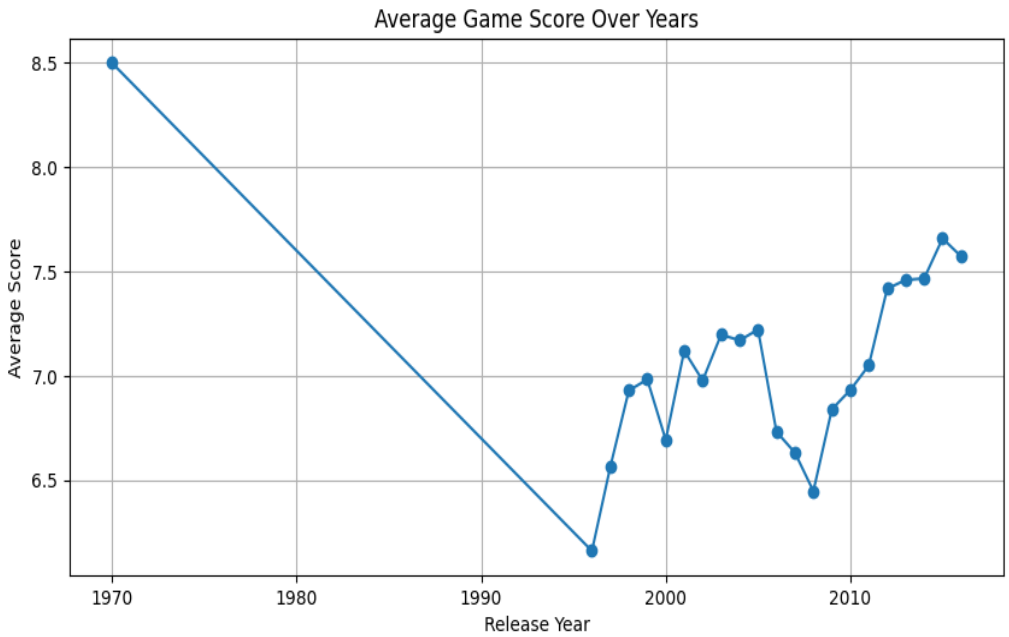
Description:

In this scenario, data is grouped by release year which is used to calculate average score per year using pandas. The results are converted into NumPy arrays for visualizations. The line graph is used to observe the trends in game quality over the years and the final plot is saved as an image for resulting purposes.

Tasks	Descrption
1	Group data by `release_year`.
2	Calculate the average score per year using Pandas.
3	Convert the resulting average scores and years into NumPy arrays.
4	Plot a line graph with `release_year` on the X-axis and average score on the Y-axis.
5	Add the title: "Average Game Score Over Years" and proper axis labels.
6	Save the graph: `plt.savefig("avg_score_trend.png")`.

Graph:

This graph describes the trend of average game review scores for different years. In this, the X-axis represents as Release Year and y-axis represents as Average Score.This graph is mainly helps to visualize the game quality is increased or not.



Conclusion:

In this scenario, the dataset was analysed how the game scores varies for different years. The data was grouped by release year and average score for each year also calculated and these values are converted into NumPy . The line graph was plotted to show trends in game quality.Finally line graph is plotted and saved it for documentation.

Scenario 3: Filtering + Bar Chart + Save (□ Medium)

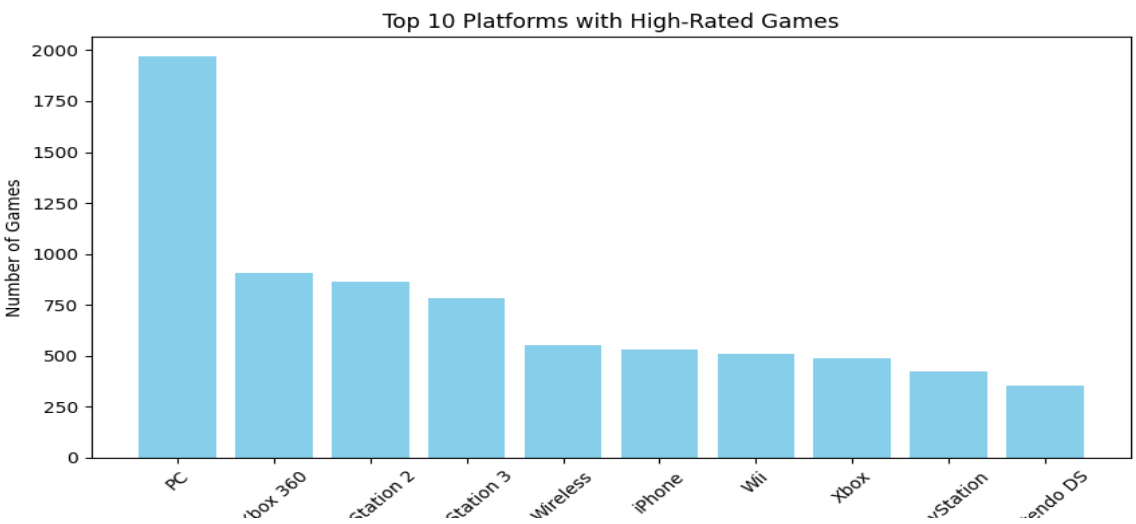
Description:

In this scenario, the dataset is mainly focussed on high rated games per platform where(score>7). The main goal is to identify the gaming platform. After filtering the dataset is grouped by platform and converting count into NumPy arrays. The top 10 platforms are selected and also visualize it by using bar chart.

Tasks	Descrption
1	Filter the dataset to include only games where `score > 7`.
2	Count the number of high-rated games per platform.
3	Select the top 10 platforms using Pandas
4	Convert the platform counts into NumPy arrays.
5	Plot a bar chart with `platform` on the X-axis and count of games on the Y-axis.
6	Plot a bar chart with `platform` on the X-axis and count of games on the Y-axis.
7	Save the graph: `plt.savefig("top_platforms_bar.png")`.

Graph:

The bar chart displays the top 10 platforms with highest no.of games which is rated above 7.The x-axis represents different platforms and y-represents as number of games.



Conclusion:

In this scenario, highlights the platforms with high-quality games. It shows that certain platforms dominates in terms of high rated titles and also strong user engagement. This bar chart visualization makes it easy to compare all platform performance.

Scenario 4: Aggregation + Pie Chart + Save (□ Medium)

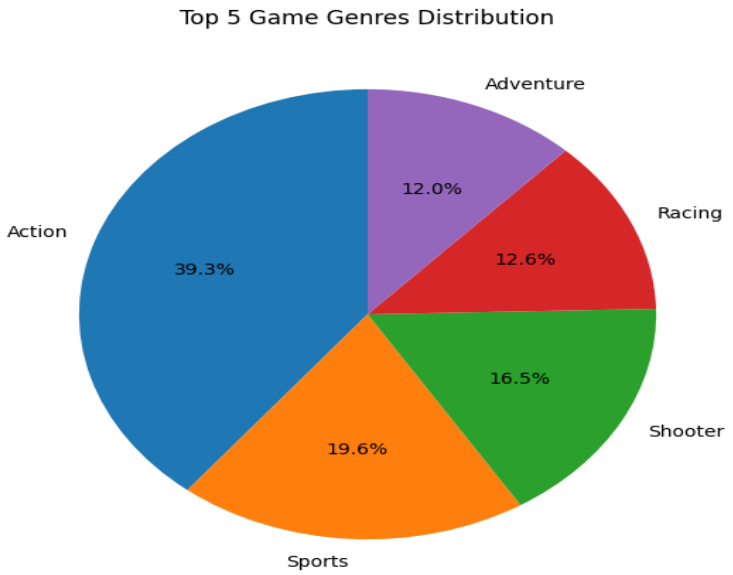
Description:

In this scenario, mainly focuses on analyses the distribution of games across different genres. The dataset is grouped by genre as labels and count as values. The top genres can be selected by visualization using pandas. percentage labels using autopct. The pie chart is used to represents proportional distribution of genres.

Tasks	Descrption
1	Count the total number of games per `genre`.
2	Select the top 5 genres using Pandas.
3	Prepare labels (genres) and values (counts) for the pie chart.
4	Plot a pie chart using `genre` as labels and `count` as values.
5	Add percentage labels using `autopct` (e.g., '%.1f%%').
6	Save the graph: `plt.savefig("genre_distribution.png")`.

Graph:

Pie chart is used to represents percentage share of top 5 genres. Each slice represents a genre. The size of slice is used to represents the number of games in that category.



Conclusion:

In this scenario, analysis reveals which genres are most popular in the dataset. A few genres dominate the majority of games, suggesting trends in player preferences and industry focus. This insight can help developers and analysts understand market demand.

Scenario 5: Advanced Analysis + Multiple Graphs (● Hard)

Description:

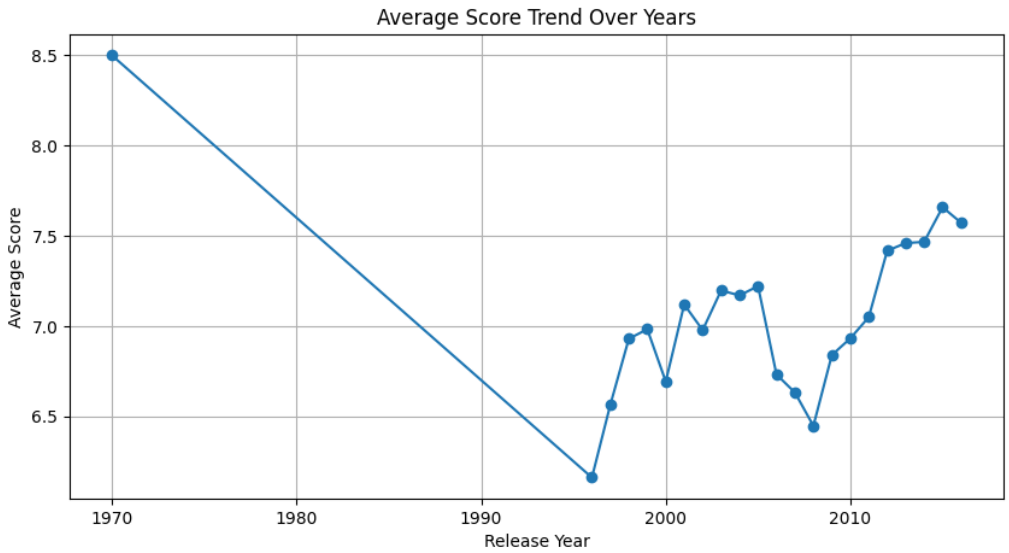
This scenario performs advanced data analysis using feature engineering, NumPy operations, and multiple visualizations. A new column (score_category) is created to classify games based on their scores. The editors-choice column is converted into numeric format for analysis. Yearly score growth is calculated using NumPy's np.diff(). Multiple visualizations—including a line graph, stacked bar chart, and histogram—are used to uncover deeper insights into review patterns.

Tasks	Descrption
1	Create a new column `score_category`: "Excellent" for score ≥ 9, "Good" for $7 \le$ score < 9, and "Average" for score < 7.
2	Convert the `editors_choice` column: 'Y' $\rightarrow 1$, 'N' $\rightarrow 0$.
3	Use NumPy's `np.diff()` to calculate the **yearly score growth** based on the average yearly scores from Scenario 2.
4	Plot a **Line Graph** showing the trend of average score per `release_year`.
5	Plot a **Stacked Bar Chart** showing the count of `score_category` per `release_year`.
6	Plot a **Histogram** showing the distribution of `score`.
7	Save the graphs: `plt.savefig("score_trend.png")`, `plt.savefig("score_category_stacked.png")`, `plt.savefig("score_distribution.png")`.
8	Identify: Which years had the highest scores, whether high scores increased over time, and if `editors_choice` correlates with high scores.

Graph:

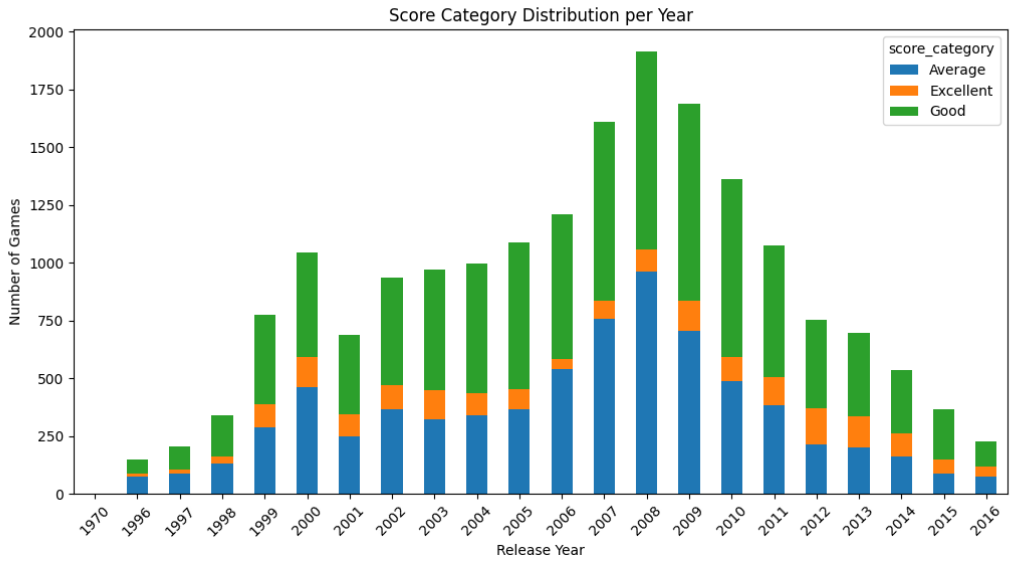
1.Line Graph (Score Growth):

Shows how average game scores change year by year. It highlights fluctuations and trends in review scores over time.



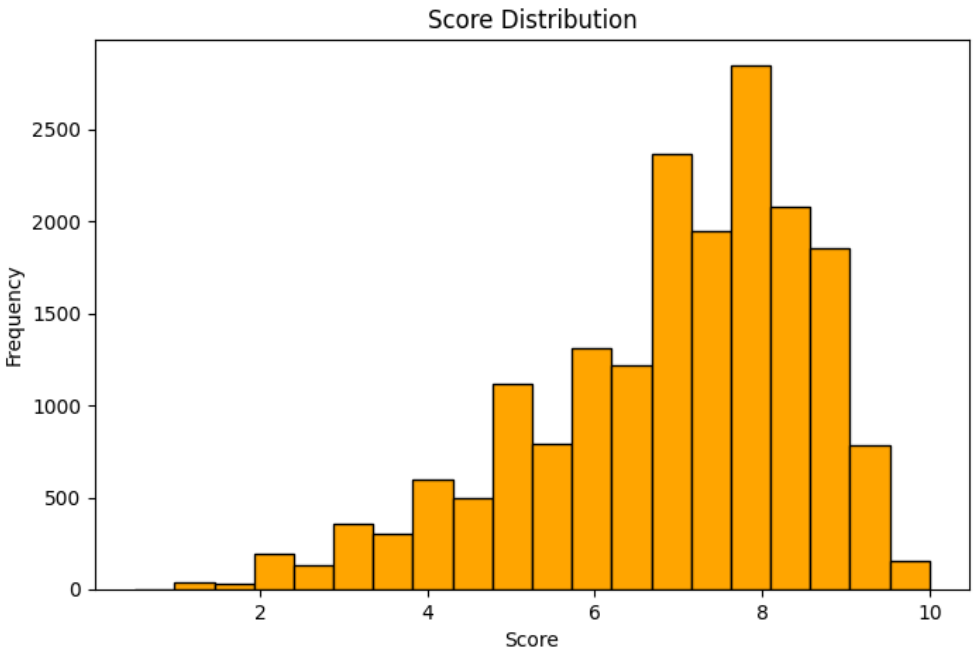
Stacked Bar Chart:

Displays the count of "Excellent", "Good", and "Average" games per year. It helps compare how score categories evolve over time.



3.Histogram:

Shows the overall distribution of game scores, indicating how frequently different score ranges occur



Conclusion:

This scenario provides deeper insights into game review patterns. Most games fall into the "Good" category, while "Excellent" games are relatively rare. The stacked bar chart shows an increase in high-rated games over time, especially after 2000. The histogram confirms that scores are concentrated between 7 and 9. Additionally, the analysis shows a strong relationship between high scores and editor's choice selections.

Final Conclusion:

This project successfully demonstrates a complete data analysis workflow using the IGN game reviews dataset. Starting from data cleaning and pre-processing, the analysis progressed through trend visualization, platform comparison, and genre distribution. Advanced techniques such as feature engineering and NumPy-based calculations were applied to extract deeper insights. The results show that game quality and volume improved significantly after the early 2000s, with certain platforms and genres dominating the industry. Overall, the project highlights how data analysis can uncover meaningful trends in the gaming data.

What you have learned

How to clean and pre-process real-world datasets using Pandas and also handling missing values using mean and mode. Where -as Performing grouping, filtering, and aggregation operations which may convert the data into NumPy arrays for numerical analysis using np.diff() to calculate trends and growth with different types of visualizations: Line charts , Bar charts ,Pie charts ,Histograms , Stacked bar charts , Saving plots for reporting and presentation ●,Interpreting data and extracting meaningful insights